Initial system

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_1^3 + x_1\left(-\sin\left(t\right) - 1\right) + x_3^2 \\ \dot{x}_3 = -x_3 + u \end{cases}$$

Series

$$x^0 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \xi_0 + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \xi_1 + \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{2} \end{bmatrix} \xi_2 + \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} \xi_{01} + \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{6} \end{bmatrix} \xi_3 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \xi_{02} + \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} \xi_{11}$$

Basis Lie elements and their coefficients

$$\begin{aligned} \mathcal{A}_1 : g_1 &= \xi_0 & v(g_1) &= \begin{bmatrix} 0 & 0 & -1 \end{bmatrix} \\ \mathcal{A}_2 : g_2 &= \xi_1 & v(g_2) &= \begin{bmatrix} 0 & 0 & -1 \end{bmatrix} \\ \mathcal{A}_3 : g_3 &= \xi_2 & v(g_3) &= \begin{bmatrix} 0 & 0 & -\frac{1}{2} \end{bmatrix} \\ g_4 &= [\xi_0, \xi_1] & v(g_4) &= \begin{bmatrix} 0 & -2 & 0 \end{bmatrix} \\ \mathcal{A}_4 : g_5 &= \xi_3 & v(g_5) &= \begin{bmatrix} 0 & 0 & -\frac{1}{6} \end{bmatrix} \\ g_6 &= [\xi_0, \xi_2] & v(g_6) &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ g_7 &= [[\xi_0, \xi_1], \xi_0] & v(g_7) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Lie elements for principal part

$$\begin{aligned} \mathcal{A}_1 : \ell_1 &= \xi_0 \\ \mathcal{A}_3 : \ell_2 &= [\xi_0, \xi_1] \\ \mathcal{A}_4 : \ell_3 &= [\xi_0, \xi_2] \end{aligned}$$

The generating set of the Right ideal

$$\begin{aligned} \mathcal{A}_2 : d_1 &= \xi_1 & v(d_1) &= \begin{bmatrix} 0 & 0 & -1 \end{bmatrix} \\ \mathcal{A}_3 : d_2 &= \xi_2 & v(d_2) &= \begin{bmatrix} 0 & 0 & -\frac{1}{2} \end{bmatrix} \\ \mathcal{A}_4 : d_3 &= \xi_3 & v(d_3) &= \begin{bmatrix} 0 & 0 & -\frac{1}{6} \end{bmatrix} \\ d_4 &= [[\xi_0, \xi_1], \xi_0] & v(d_4) &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Principal part

$$\mathcal{A}_1 : \tilde{\ell}_1 = \xi_0$$
$$\mathcal{A}_3 : \tilde{\ell}_2 = \xi_{01}$$
$$\mathcal{A}_4 : \tilde{\ell}_3 = \xi_{02}$$

Approximating system

$$\begin{cases} \dot{x}_1 = -u \\ \dot{x}_2 = -tx_1u \\ \dot{x}_3 = -t^2x_1u \end{cases}$$

Stationary approximating system

$$\begin{cases} \dot{x}_1 = -u \\ \dot{x}_2 = -\dfrac{x_1^2}{2} \\ \dot{x}_3 = -2x_2 \end{cases}$$